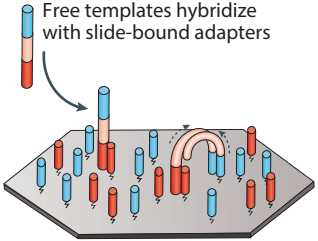
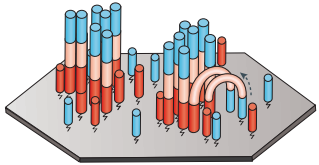
Solid-phase bridge amplification (Illumina)

Template binding
Free templates hybridize
with slide-bound adapters

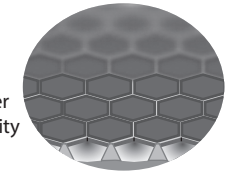


Bridge amplification:
Distal ends of hybridizedtemplates
interact with nearby primers
where amplification can take place



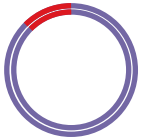
Cluster generation **:**
After several rounds of amplification,
100–200 million clonal clusters are formed

Patterned flow cell:
Microwells on flow cell direct cluster
generation, increasing cluster density



In-solution DNA nanoball generation (Complete Genomics (BGI))

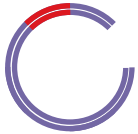




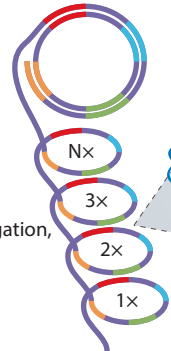
Cleavage Circular
DNA templates are
cleaved downstream
of the adapter sequence



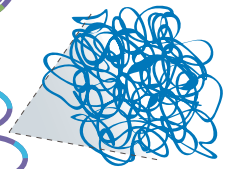
Adapter ligation
One set of adapters is
ligated to either end
of a DNA template, followed
by template circularization

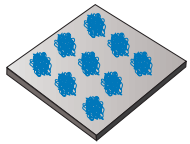


Iterative ligation
Three additional rounds of ligation,
circularization and cleavage
generate a circular template
with four different adapters



Rolling circle amplification
Circular templates are amplified to
generated long concatamers,
called DNA nanoballs;
intermolecular interactions keep the
nanoballs cohesive and separate in solution





Hybridization
DNA nanoballs are immobilized
on a patterned flow cell